

VSA Megacode (ACLS 2020)

Section 1: Case Summary

Scenario Title:	Vital Signs Absent (VSA) Megacode
Keywords:	Resuscitation, CPR, Cardiac, RCDP
Brief Description of Case:	A 54-year-old male police officer presents to the ED complaining of chest pain for two hours that started during his weekend hockey game. He is feeling dizzy and short of breath upon presentation. He will have a VT arrest as he is placed on the monitor. He will require two shocks and rounds of CPR before he has ROSC. He will then lose his pulse again while the team is trying to initiate post-arrest care; This will happen several times. Finally, the team will maintain ROSC. An ECG will show a STEMI and the team will need to call for emergent PCI.

Goals and Objectives	
Educational Goal:	To provide learners with the opportunity to manage a prolonged resuscitation that requires strong resource allocation skills.
Objectives: (Medical and CRM)	1. Provide quality ACLS care, including: a. Expedient defibrillation for a shockable rhythm b. Appropriate use of ACLS medications c. Use of ETCO₂ to guide resuscitation d. Minimizing pulse checks e. 30:2 ratio of compressions to breaths prior to intubation 2. Initiate appropriate post-arrest care, including PCI. 3. Communicate clearly and lead team through a lengthy resuscitation. 4. Recognize compression fatigue in team members and find ways to engage or relieve fatigued members
EPAs Assessed:	

Learners, Setting and Personnel					
Target Learners:	☒ Junior Learners		☐ Senior Learners		☐ Staff
	☒ Physicians	☐ Nurses	☐ RTs	☐ Inter-professional	
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Sim Actors: 1-2				
	Sim Techs: 1				

Scenario Development	
Date of Development:	2015
Scenario Developer(s):	Dr. Cheryl French
Affiliations/Institutions(s):	University of Manitoba
Contact E-mail:	
Last Revision Date:	Jan 2023
Revised By:	Dr. Chris Bennett Heyd
Version Number:	1



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Richard Green		Age: 54		Gender: Male	
Weight: 80kg					
Presenting complaint: Chest pain					
Temp: 36.7		HR: 74		BP: 105/65	
RR: 16		O ₂ Sat: 99%		FiO ₂ : RA	
Cap glucose: 6.7		GCS: 15			
Triage note: Played his usual weekend hockey game about 2 hours ago. Developed chest pain during the hockey game. It got better after he stopped and rested for a bit, but it hasn't really gone away.					
Allergies: none					
Past Medical History: none			Current Medications: none		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
Works as a police officer No history of chest pain No previous cardiac testing Non-smoker Unknown family history	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal heart sound, normal pulses	Neuro: Normal
Resp: Normal	Head & Neck: Normal
Abdo: Normal	MSK/skin: Diaphoresis
Other: None	



Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin: Adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Crash cart CPR board Airway cart
C. Required Medications
Crash cart/ACLS medications RSI medications
D. Moulage
Diaphoresis (optional)
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> Feels lightheaded on arrival Chest pain (heaviness) x2 hours. Retrosternal, radiates to L arm Feels SOB from walking into department Mild nausea

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Bedside RN(s)	Helpful and skilled May use multiple nurses for: medications, charting, CPR
Resp therapist (optional)	Helpful and skilled For airway management during the resuscitation

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 74/min BP: 100/60 RR: 12/min O ₂ Sat: 96% RA T: 36.7 °C GCS: 15	<i>Alert. Diaphoretic. Complaining of increased CP, SOB, dizzy.</i>	<u>Expected Learner Actions</u> ☐ IV, O ₂ , monitors ☐ ECG (not given) ☐ Cardiac blood work ☐ History and Physical ☐ ASA 160mg PO chew ☐ Nitro spray	<u>Modifiers</u> <i>Changes to patient condition based on learner actions</i> Nitro spray → BP 80/40 <u>Triggers</u> <i>For progression to next state</i> - 2min → 2. VT arrest	
2. VT arrest Rhythm: VT HR: 180 BP: undetectable O ₂ Sat: 74% ETCO ₂ : 10	<i>Patient unconscious and pulseless.</i>	<u>Expected Learner Actions</u> ☐ Immediate defibrillation ☐ High quality CPR ☐ Confirm IV access ☐ Initiate capnography ☐ Compressions:BVM at 30:2 ☐ Prepare advanced airway	<u>Modifiers</u> - 1 st shock → stays in VT - Quality CPR: ETCO ₂ → 16 - Poor CPR: ETCO ₂ → 8 <u>Triggers</u> - 2 nd shock → 3. Brief ROSC	Consider ETT or LMA for airway management Do not give epi or amio before 2 nd shock per ACLS
3. Brief ROSC Rhythm: Junctional HR: 55 BP: 85/45 O ₂ Sat: 95% ETCO ₂ : 40	<i>Patient opens eyes to voice.</i>	<u>Expected Learner Actions</u> ☐ Recheck vital signs ☐ Ask for ECG (not given) ☐ Fluid bolus ☐ Norepi or epi infusion ☐ Consider amiodarone	<u>Modifiers</u> <u>Triggers</u> - 1 min into state 1st time → 4. VF arrest - 1 min into state 2 nd time → 5. PEA arrest	



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4. VF arrest Rhythm: VF HR:180 BP: undetectable O ₂ Sat: 68% ETCO ₂ : 10	<i>Patient unconscious and pulseless.</i>	<u>Expected Learner Actions</u> ☐ Immediate defibrillation ☐ High quality CPR ☐ Epinephrine 1amp q3min ☐ Amiodarone 300mg ☐ Advanced airway ☐ Monitor capnography	<u>Modifiers</u> - 1 st shock → stays in VF - Quality CPR: ETCO ₂ → 16 - Poor CPR: ETCO ₂ → 8 <u>Triggers</u> - 2 nd shock → 3. Brief ROSC	Consider ETT or LMA for airway management
5. PEA arrest Rhythm: slow PEA HR: 20 BP: undetectable O ₂ Sat: 68% ETCO ₂ : 10	<i>Patient unconscious and pulseless.</i>	<u>Expected Learner Actions</u> ☐ High quality CPR ☐ Epinephrine 1amp q3min ☐ Consider HCO ₃ and CaCl ₂ ☐ Advanced airway (if not yet) ☐ Monitor capnography	<u>Modifiers</u> - Quality CPR: ETCO ₂ → 16 - Poor CPR: ETCO ₂ → 8 <u>Triggers</u> - 3 rounds or 15 min → 6. ROSC	
6. ROSC Rhythm: Sinus/STEMI HR: 55 BP: 85/45 O ₂ Sat: 96% RR:12 (vented) ETCO ₂ : 40	<i>Patient nonresponsive but has pulse</i>	<u>Expected Learner Actions</u> ☐ ECG ☐ Cap sugar (6.2) ☐ Norepi or epi infusion ☐ OG for clopidogrel/ticagrelor ☐ IV Heparin ☐ Call interventional cardiology ☐ Consider cooling	<u>Modifiers</u> - Vasopressor started → BP 95/65 <u>Triggers</u> - 20 min or cardiology called → End Case	

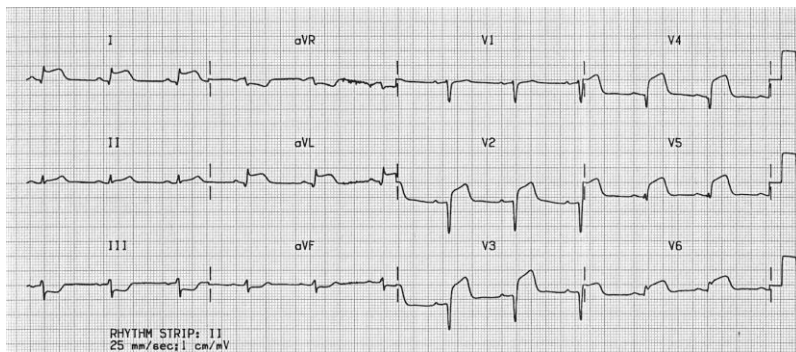
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Appendix A: Laboratory Results

None given during case	
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



(ECG source: <http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/10/anterolateral.jpg>)

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Sample Questions for Debriefing

- 1) How did it feel to have your resuscitation intermittently produce ROSC?
- 2) This was a long and frustrating case. Did you consider the fatigue of team members? What was your strategy to address fatigue?
- 3) Were you using ETCO₂ to assess compression quality and ROSC? How is ETCO₂ helpful?
- 4) How and when do you initiate therapeutic hypothermia after a cardiac arrest? When is it contraindicated?

Key Moments

- Recognition of loss of pulse, initiation of ACLS
- Recognition of ROSC and then repeated pulse loss
- Recognition of STEMI and calling cath lab

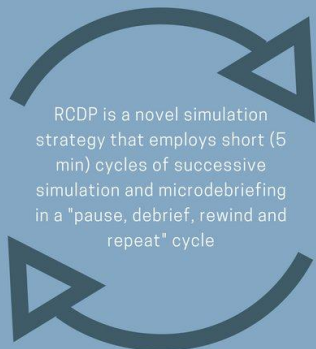
Consider Using Rapid Cycle Deliberate Practice

- See infographic on the following page

RAPID CYCLE DELIBERATE PRACTICE

A primer for simulationists

Alia Dharamsi (@alia_dh) #SHRED Fellow, University of Toronto



RCDP motivates a team to work together to achieve best practice standards



RCDP propels learners forward in their resuscitation skills, using repetition to transition from uncertainty to mastery

DO IT RIGHT

In this model, learners have multiple opportunities to try, learn, and immediately implement new skills in order to optimize patient outcomes.



FIRST-FIVE MINUTES

Much of the research in RCDP involves refining the "first-five minutes" of a resuscitation. Metric-based targets (ie. initiation of compressions, time to defibrillation, and minimizing peri-shock pauses) are particularly amenable to RCDP as they are tangible targets.

COACHING IS KEY



This process relies upon effective real-time and directed feedback. Debrief time is abbreviated, thus feedback needs to be goal-oriented and highlight key action items for the next iteration. Emphasis should be placed on modifiable actions.



RCDP IMPROVES LEARNERS'

RETENTION of resuscitation principles

KNOWLEDGE of algorithms

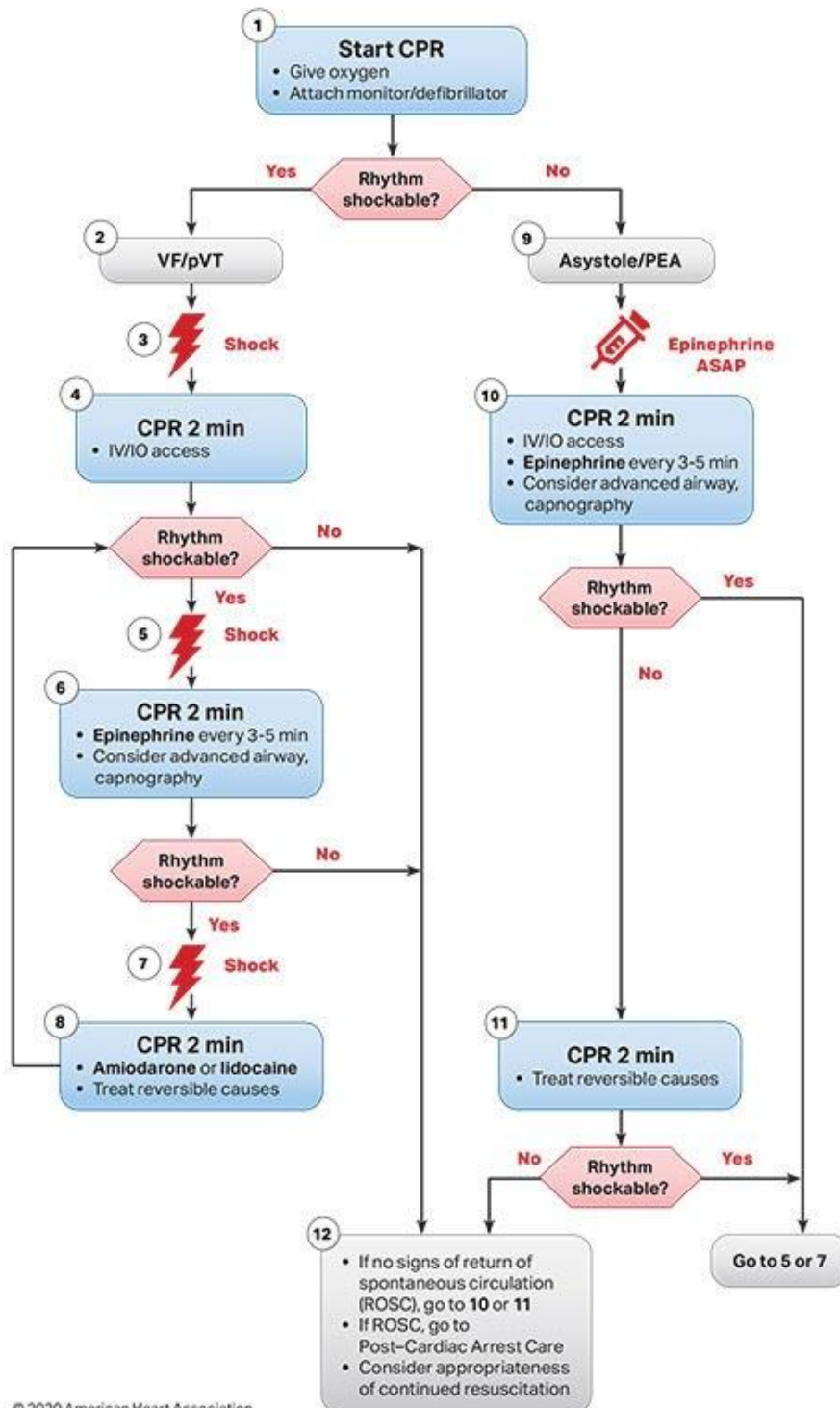
CONFIDENCE in team roles

Resources and References:

1. Taras J, Everett T. Rapid Cycle Deliberate Practice in Medical Education - a Systematic Review. *Cureus*. 2017;9(4):e1180. Epub 2017/04/19. doi: 10.7759/cureus.1180. PubMed PMID: 29540142; PubMed Central PMCID: PMC5411688.
2. Lemke DS, Fielder EK, Hsu DC, Doughty CB. Improved Team Performance During Pediatric Resuscitations After Rapid Cycle Deliberate Practice Compared With Traditional Debriefing: A Pilot Study. *Pediatr Emerg Care*. 2016. Epub 2016/10/06. doi: 10.1097/PEC.0000000000000940. PubMed PMID: 27741771.
3. Hunt EA, Duval-Arnould JM, Nelson-McMillan KL, Bradshaw JH, Diener-West M, Perretta JS, et al. Pediatric resident resuscitation skills improve after "rapid cycle deliberate practice" training. *Resuscitation*. 2014;85(7):945-51. Epub 2014/03/04. doi: 10.1016/j.resuscitation.2014.02.025. PubMed PMID: 24607871.

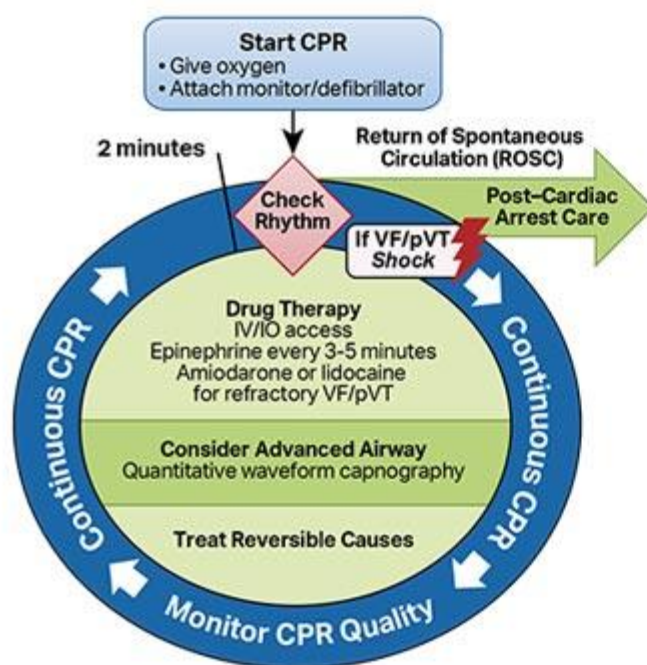
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Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)



CPR Quality
- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil. - Minimize interruptions in compressions. - Avoid excessive ventilation. - Change compressor every 2 minutes, or sooner if fatigued. - If no advanced airway, 30:2 compression-ventilation ratio. - Quantitative waveform capnography - If PETCO₂ is low or decreasing, reassess CPR quality.
Shock Energy for Defibrillation
Biphasic: Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered. Monophasic: 360 J
Drug Therapy
Epinephrine IV/IO dose: 1 mg every 3-5 minutes Amiodarone IV/IO dose: First dose: 300 mg bolus. Second dose: 150 mg. Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.
Advanced Airway
- Endotracheal intubation or supraglottic advanced airway - Waveform capnography or capnometry to confirm and monitor ET tube placement - Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions
Return of Spontaneous Circulation (ROSC)
- Pulse and blood pressure - Abrupt sustained increase in PETCO₂ (typically ≥40 mm Hg) - Spontaneous arterial pressure waves with intra-arterial monitoring
Reversible Causes
- Hypovolemia - Hypoxia - Hydrogen ion (acidosis) - Hypo-/hyperkalemia - Hypothermia - Tension pneumothorax - Tamponade, cardiac - Toxins - Thrombosis, pulmonary - Thrombosis, coronary

Adult Cardiac Arrest Circular Algorithm



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- Hypothermia
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- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

References

1. American Heart Association. 2020. ACLS Algorithms. <https://cpr.heart.org/en/resuscitation-science/cpr-and-ecc-guidelines/algorithms>